

1. System Overview

Problem Definition

Customer support systems often rely on static FAQs or manual responses, leading to delayed or inaccurate answers. Users expect quick, contextual, and intelligent responses.

Proposed Solution

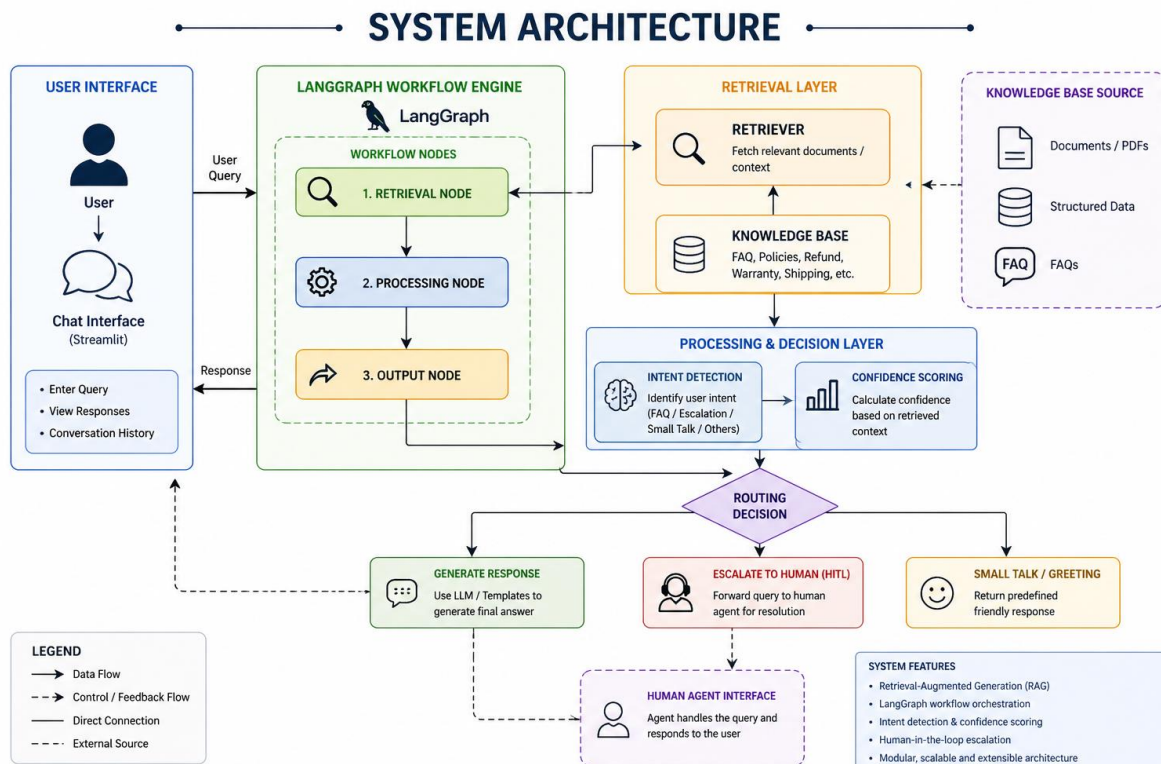
This project implements a Retrieval-Augmented Generation (RAG) based Customer Support Assistant that:

- Retrieves relevant information from a knowledge base
- Generates contextual responses
- Uses a graph-based workflow (LangGraph) for control logic
- Supports Human-in-the-Loop (HITL) escalation for complex queries

Scope of System

- Process customer support queries (refund, warranty, shipping)
- Provide automated responses
- Escalate to human when required
- Provide web-based chat interface

2. System Architecture



3. Component Description

3.1 User Interface

- Chat-based interaction
- Displays conversation history

3.2 Document Loader

- Loads knowledge base
- Extendable to PDFs

3.3 Chunking Strategy

- Splits data into smaller chunks (future)

3.4 Embedding Model

- Converts text into vectors (future)

3.5 Vector Store

- Stores embeddings (ChromaDB future)

3.6 Retriever

- Matches query with knowledge base

3.7 LLM Layer

- Generates responses (simulated)

3.8 Workflow Engine (LangGraph)

- Controls execution flow
- Nodes: Retrieval → Processing → Output

3.9 Routing Layer

- Decision making using:
 - Intent detection
 - Confidence score

3.10 HITL Module

- Handles escalation cases

4. Data Flow

1. User inputs query
2. Query sent to workflow
3. Context retrieved
4. Processing evaluates
5. Routing decision made
6. Output generated

7. 5. Technology Choices

- Streamlit → UI
- LangGraph → Workflow
- Python → Development
- ChromaDB → Future vector DB

6. Scalability

- Add vector database
- Handle large documents
- Cloud deployment
- Caching

7. System Advantages

- Modular design
- Scalable architecture
- Intelligent decision making
- Supports escalation